



Indian Railways' Strategic Vision to Serve Ports:

A Case of Gandhidham Area

1. Introduction

"There is a lot of business opportunity available here", said the Area Railway Manager, Gandhidham, while explaining the potential of the Area. "The two ports of Kandla and Mundra have significant growth plans. The non-pipeline cargo at Kandla is expected to increase from 18 million tons in 2003-04 to 21 million tons in 2006-07 and 31 million tons in 2011-12. The non-pipeline cargo at Mundra is expected to increase from 4.5 million tons in 2003-04 to 41 million tons in 2011-12."

"In terms of rail traffic, Gandhidham Area dealt with a total of 4,456 loaded *rakes* in 2003-04. We expect the traffic to grow to a possible 7,226 rake loads by 2006-07 and 12,453 rake loads by 2011-12. Therefore, it is important for railways to recognize the potential, right now, so that adequate infrastructure and systems are put into place to take advantage of the growth possibilities. Otherwise, we may lose this business to other modes of transport, including road."

Gandhidham Area: Average Number of Rakes per Month 2003-04

Type of Rake	Received		Sent	
	Loaded	Empty	Loaded	Empty
BATL	8.4	0.0	0.0	8.4
BOXN	3.5	11.0	14.1	0.4
BTPN	12.8	0.0	0.0	12.8
CONT	9.7	0.0	7.0	2.7
FO Tank	6.9	0.0	1.2	5.7
JUMBO	161.8	5.9	135.1	32.8
LPG Tank	0.0	6.4	6.4	0.0
V-OIL Tank	0.0	7.7	7.7	0.0
Total	203.1	31.0	171.5	62.8

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Teaching Material of the IIM, Ahmedabad and RSC, Vadodara, is prepared as a basis for class discussion. Cases are not designed to present illustrations of either correct or incorrect handling of administrative problems.

He further added, "The traffic flow pattern in this Area is very interesting. For example, in the year 2002-03, Mundra port received 391 rakes of wheat and rice for export (out of 394 total unloaded) and they loaded 287 rakes of imported coal (out of 365 total loaded). Prima facie, it appears that traffic flow is fairly matched, however, there are times when I have goods to carry but I dispatch empty rakes to other divisions!"

2. Indian Railways' Operations at Gandhidham

Gandhidham was a major junction, catering to Kutch region of Gujarat. It had proximity to the two ports of Kandla and Mundra. **Exhibit 1** provides a general and historical perspective of Gandhidham Area, which formed a part of the Ahmedabad Division of the Western Railway (WR).

Rail connectivity was available to Delhi and North India, both by Broad Gauge (BG) and Meter Gauge (MG) and to Ahmedabad, Mumbai, and the rest of the country by BG. **Exhibit 2** shows the railway network in the region. Its geographical location gave significant advantage to Gandhidham, from the point of view of railways, as it brought in substantial portion of goods income. There were 14 major freight handling stations/sidings in the Gandhidham Area. **Exhibit 3** shows the layout of Gandhidham Area.

The main outward commodities were salt, fertilizers, POL, coal, edible oils, HR coils, containers etc. The main inward commodities were food grains, iron coils, DOC, hay, military store, POL, ammonia, containers etc. **Exhibit 4** gives stationwise loading and unloading points in the Gandhidham Area.

3. Kandla Port

Kandla port was set up in 1952 and declared as a major port in 1955. It was primarily set up as a replacement to Karachi port, which became part of Pakistan after partition from India in 1947. With time, it had evolved as one the most important ports of India. It was a natural harbour situated on the Kandla Creek in Gujarat.

Kandla port was one of the 12 major ports of India, administered by the central government. **Exhibit 5** shows geographical location of the major ports of India.

Kandla port was connected with the major cities and industrial centres in India by National Highway 8A (NH8A), apart from the railway system. As per the port authorities, "Kandla port meets the requirements of hinterland spread over one million square kilometres catering to the needs of whole of North India including Jammu & Kashmir, Punjab, Himachal Pradesh, Haryana, Rajasthan, New Delhi and Gujarat. It deals in food grains, fertilizers and fertilizer's raw materials, timber logs, metal scrap, petroleum and other liquids (POL), chemicals and hazardous cargo, edible oil, iron ore, coal, containers, granites etc."

Kandla port had 11 dry cargo berths, one steel floating dry dock, three jetties for small vessels, one maintenance jetty, six oil jetties, and two Single Buoy Moorings (SBMs) at Vadinar for crude oil imports.

In 2002-03, these 12 major ports accounted for 313 million tonnes (mt), which was about 75% of total port traffic (the balance being accounted for by the minor ports under the charge of various state governments).

In 2002-03, Kandla handled 41 mt, forming 13% share of the major ports. In 2003-04, Kandla handled 42 mt showing a growth of 2% over the previous year. Out of 2003-04 traffic, crude oil was 22 mt, liquid bulk was 6 mt, and dry bulk cargo was 14 mt. The crude oil imports were evacuated by pipeline to inland refineries at Vadodara, Mathura, and Panipat.

Out of the 6 mt of liquid bulk, 2 mt were raw materials for the IFFCO fertilizer plant at Kandla. About 2 mt were POL, most of which were imports, though about 0.3 mt were also exported as coastal traffic. Other liquids like edible oil constituted 2 mt of traffic. Out of the 4 mt of liquid traffic that moved from and to the hinterland the POL for coastal exports was handled by railways as also a small share of the imported POL. Over 5 mt of the liquid bulk were imports.

Out of the 14 mt of dry bulk, 9.7 mt were exports. Of the exports, 4.7 mt were food grains, 1.6 mt were de-oiled cake (DOC), 1.6 mt of containerised traffic, and 0.9 mt of salt. Out of the imports, 1.4 mt were timber, 0.8 mt of containerised traffic, and 0.4 mt were iron scrap.

Exhibit 6 and **Exhibit 7** give the rail share of total export and import traffic respectively at Kandla port for the past five years. On the export front, the quantity handled at the port more than doubled between 1999-00 and 2000-01, primarily due to foodgrains. The rail share also went up significantly. On the import front, there was a significant drop between 1999-00 and 2000-01, which level has more or less continued until 2003-04. This was primarily due to the drop in POL imports, followed by edible oil and DAP/UREA. POL imports fell due to the opening of Reliance refinery at Sikka. Edible oil shifted to Mundra. The rail share of imports also dropped significantly between 1999-00 and 2000-01 and has shown a declining trend until 2003-04.

To assess the real hinterland reach for Kandla port, an analysis of hinterland distance profile of cargo handled at Kandla port was attempted. Since, data was not readily available for dry bulk, therefore, analysis was carried out with liquid bulk cargo data available only for the recent months (provided by the Department of Customs and Excise to the port). **Exhibit 8** shows the hinterland distance profile for top ten liquid bulk commodities handled at the port for the months of November 2003 and February 2004.

4. Mundra Port

The Mundra port was a joint venture between the Government of Gujarat and the Rs. 45 billion Adani Group, operated as a company called Gujarat Adani Port Limited (GAPL). It started functioning as a modern deep draft port during 1998-1999. The structure of ownership of Mundra Port is given in **Exhibit 9**.

According to the website of Mundra Port (www.mundraport.com), "It is a multi-purpose all weather port that can receive dry, liquid cargo and container ships up to 1,30,000 dead weight ton (DWT) with a clear and deep water approach and a minimum natural depth of 18 m at any state of the tide."

Gujarat government had identified the port sector as one of the prime areas for development of infrastructure of the state. The government also viewed the ports as a key facilitator of overall development in the state. To implement this vision, in 1995, the government formulated a port policy. The main feature of this policy was to move towards privatisation of the ports and allied services to achieve global standards, and to drive industrial development along the coast by leveraging the ports.

The Gujarat Maritime Board (GMB), an autonomous state board set up in 1982, was entrusted with developing and operating ports in the state. GMB was working towards achieving the vision of the government. Apart from GMB's own ports, GMB had facilitated development of captive ports/jetties, joint sector ports (like GAPL), and private ports. There were 40 ports under GMB, of which 12 carried cargo while the remaining were primarily fishing harbours. **Exhibit 10** shows geographical location of ports under GMB.

India had a total of 140 minor ports, under the administration of various maritime states. The GMB ports were thus classified as minor ports. In 2002-03, these 140 minor ports accounted for 105 mt, which was about 25% of total port traffic. Of these, the GMB ports accounted for 84 mt. Sikka, one of the captive 'minor' ports under GMB, was set to become the largest port in the country with a total traffic of over 45 mt in 2003-04.

In 2002-03, Mundra handled over 4.2 mt, forming 4% share of the minor ports. In 2003-04 (until February), Mundra handled over 4.1 mt. In 2003-04 (until February), dry cargo handled was 3.1 mt – out of which, minerals were 0.6 mt, wheat, coal, and steel above 0.5 mt each, and fertilizers and clinker contributed another 0.3 mt each. In the same period, liquid cargo handled was close to 1 mt – out of which, edible oils was about 0.6 mt and the rest were chemicals and POL. **Exhibit 11** gives the annual cargowise details since the inception of the port.

A 56 km long siding to connect the Mundra port to the nearest existing railway station was constructed through a Special Purpose Vehicle (SPV). The siding was opened for traffic during 2001-02. **Exhibit 12** and **Exhibit 13** give the rail cargo handled at Mundra, both in terms of rakes and tonnage, for export and import traffic, respectively.

In 2003-04 (as per IR data, and including March), export cargo was 2.2 mt, out of which, wheat was about 0.7 mt, bauxite was about 0.5 mt, clinker was 0.35 mt, bentonite was about 0.28 mt (bulk and bagged), steel pipes and DOC were above 0.1 mt each. The import cargo, for the same period, was 2.0 mt, out of which, coal was about 0.7 mt, edible oils were over 0.5 mt, steel plates/HR coils were about 0.4 mt, DAP/urea/MOP contributed another 0.37 mt. **Exhibit 14** and **Exhibit 15** give the rail share of total export and import traffic respectively at Mundra port for the past three years.

5. Traffic Potential for Railways

Due to the encouragement provided by the industrialization policy of the government of Gujarat, industries had come up near the port, thus requiring a short lead movement.

For example, the IFFCO fertilizer plant was situated adjacent to Kandla port, and moved its raw material from the port to the plant by pipeline. Similarly, Sanghi Cement, Jindal Steel Tubes Limited, and Adani-Wilmar Limited were located close to Mundra port, sourcing requisite raw materials from the port. However, in all these industries, the finished products moved into the hinterland and could potentially be moved by railway.

In the case of IFFCO, most of the finished fertilizer was moved by rail in Jumbos. In the case of Sanghi Cement and Adani-Wilmar Limited, some of the finished product moved by rail, also by Jumbos.

Mr. Singh of Adani-Wilmar reasoned the low usage of rail movement as, "Our produce is transported in cartons. In a wagon, I can load at the most six layers of cartons containing poly-packs of oil, one above the other. When we tried loading more than six, many cartons

suffered damage. The cartons I can load weigh 22 tons only in one wagon, while I have to pay for minimum load of 26 tons per wagon. Therefore I confine myself to sending oil by rail for North-Eastern region only, where road movement is not as safe as it should be.”

Sanghi Cement had their own captive jetty at Jakhau, which they were increasingly using for coastal movement for sourcing of coal, and distribution of clinker and finished cement.

Jindal Steel Tubes Limited was an export oriented unit, located about 25 kms from Mundra port, for which the steel plates (raw material) were imported and the tubes (finished product) were exported at Mundra.

Salt farming, bentonite and alumina mining were carried out within a radius of 200 kms from the ports and exported. Significant quantities of salt were also moved from the area into the hinterland for domestic consumption, mostly in Jumbos.

In terms of long lead traffic, food grain, coal, POL, timber, metal scrap, etc. were possibilities for movement by railways. Even now, a significant share of food grains (in Jumbos) and coal (in BOX-N) for power plants moved by railways. As per the trade, more could be moved if wagon availability was not a constraint. Wagon designs which permitted greater flexibility of loading different commodities could help.

Most of the POL for coastal exports moved by railways. Leveraging on these incoming tank wagons, some of the POL imports were moved by railways.

In the case of timber and metal scrap, both of which required open wagons like BOX-N, the traffic moved by road in the trucks returning after unloading food grain for export from the hinterland. In terms of destination profile, large volumes of these commodities did move to single destinations like steel processing plants at Ludhiana and Kosi Kalan, and timber processing for furniture around Delhi. The trade moved these goods by road, primarily, due to the lower tariffs. The road tariffs were negotiable and varied as a function of demand and supply for trucks at Gandhidham. A typical rate to Delhi area for a standard nine ton truck was in the range of Rs 9000 to Rs 12000. Exhibit 16 gives the rail tariff for the major commodities to different destinations. The industry perceived the rail rates as being non-negotiable. Further, rates were charged based on minimum loading requirements. The first mile and last mile costs also added to the total cost of transporting by train.

The timber was received as logs, imported from Myanmar and Indonesia. It was then predominantly sawn into loadable sizes at Gandhidham (as per the requirement of customer) and then sent by road to the hinterland to Northern India. Sawn timber could also move in Jumbos.

Crude edible oil was converted into value added products ranging from oils to vanaspati. This industry was expected to grow due to excise duty breaks provided by the government of Gujarat in earthquake affected areas. Such concessions were available for five years for any industry that would be set up till 2007.

6. Future Possibilities

The Kandla Port Authority had drawn up plans to construct two more berths on a Build-Operate-Transfer (BOT) basis to handle container traffic. The eleventh and the twelfth berths (as they would be numbered) would have a combined quay length of 545 meters and a depth of 12.5 meters alongside the berths. Berth number 11 was already available for use.

Berth number 12, was under development and was expected to be ready for use by 31st October 2005. The port had projected a traffic of 1.5 lakh TEUs in 2005-06 and 4.5 lakh TEUs in 2009-10. Some analysts felt that the Kandla port's projections may be on the higher side, going by the movement of containers in that region.

Adjacent to these berths, inside the custom bonded area, 40 hectares of land were to be developed as a container stack yard. Container handling operations at berth number 12 were to be carried out with at least two new Rail Mounted Quay Cranes (RMQCs). The other berths where container ships would visit necessarily had to use ship based gear for loading and unloading.

Plans were also being made with a concept of impounded jetties to reduce the vulnerability to tidal variations that was present in the existing berths and to accommodate deeper draft vessels. This facility would be located at Tuna Buoy, some distance away from the present location. Initial plans were for four jetties with scope for expansion as per increasing requirements.

Mundra port was planning to expand aggressively with more finger jetties for dry and liquid bulk. They were also planning investments in SBMs, jointly with one of the petroleum companies. The growth vision for Mundra Port as articulated by Mr. Gautam Adani, Chairman, Adani Group was, "To evolve as an international calibre business port on the foundations of technological excellence, state-of the-art back up infrastructure and logistics, and a competitive customer focus. We are committed to making Mundra port into a maritime masterpiece".

The traffic projections of these two ports as consolidated by the Area Railway Manager, Gandhidham is given in **Exhibit 17**. The existing rake handling capacity at various stations in this Area is given in **Exhibit 18**. The present line capacity of the expected bottleneck section of Gandhidham to Maliya is given in **Exhibit 19**.

Gandhidham Area was also expected to be affected by the overall developments in various GMB ports. GMB's vision for the year 2020 is given in **Exhibit 20**, along with major growth commodities in **Exhibit 21**.

In response to such projections, plans were being proposed by the Gandhidham Area for future investments. **Exhibit 22** gives the status of various works in progress or planned in Gandhidham and Kandla goods shed. **Exhibit 23** gives the nature of future projects in the Gandhidham Area. For example: (i) Item 3 – double line between GIM BG yard and KDLP section was to cater for expected phosphoric acid traffic for IFFCO plant and anticipated rise in container traffic at Kandla port, (ii) Item 5 – conversion of Gandhidham block station into full fledged block station with two loops (one loop as goods shed) was to improve junction point facilities and augment goods handling capacity, (iii) Item 7 – bypass chord to SIOB-LKZ during conversion of SIOB-PNU section was to provide an alternative route for traffic originating in Saurashtra region to be moved via Bhildi- Palanpur, (iv) Item 11 – Expansion of GIM yard, and conversion of existing MG yard into BG yard was due to MG no more being used and expanding BG yard, and (v) Item 16 – covered goods shed at GIM and KDLP was proposed on account of an MOU with Central Warehousing Corporation (CWC) who had agreed to construct a goods shed.

The Area Railway Manager Gandhidham, further said, "The most important project was the sanctioned gauge conversion from Gandhidham to Palanpur, which would provide a

shorter BG connectivity to North India, and more importantly avoiding the congested Ahmedabad area. This project would provide a double line capacity from Gandhidham to Samakhiyali, which also carried the traffic towards Ahmedabad. Given the increasing traffic to Mundra port, the need for a reversal at Gandhidham was to be avoided by the construction of a bypass line. This, coupled with other improvements in the Area, was expected to provide opportunities for traffic growth. It would also be important to see where else in the IR system, improvements would be required for dealing with the growing traffic.”

Glossary

BATL: Eight wheeler tank wagons for moving liquid Ammonia.

BOT: Build Operate Transfer. The preferred form of public private partnership in infrastructure development and services.

BOXN: Eight wheeler air braked open wagons.

BTPN: Eight wheeler air braked tank wagons.

CONT: Container wagons.

DWT: Commodity carrying capacity of a ship.

FO TANK: Furnace Oil Four wheeler tank wagons.

JUMBO: A rake comprising Eight wheeler covered wagons.

LPG Tank: Four/Eight wheeler tank wagons for moving Liquified Petroleum Gas.

Rake: The unit of freight traffic movement in Indian Railways is termed as a rake. A rake consists of specified number of wagons that vary depending on the capacity of the motive power as also the commodity carried in the rake. For example, coal is carried in rakes, which comprise of 58 wagons of a particular type.

SBM: Single Buoy Mooring. An off shore pipeline terminal, typically for liquid bulk loading/unloading onto ships. Being off shore, it permits deep draft ships.

V-OIL Tank: Four wheeler tank wagons for moving vegetable oil.

Exhibit 1

Gandhidham Area – A Perspective

The Gandhidham Area of Ahmedabad Division, Western Railway (WR) covered the Palanpur–Gandhidham–Kandla Port, and Gandhidham–New Bhuj–Naliya sections. These were mainly in the Kutch district of Gujarat which itself covered a quarter of the area of the state. The remaining part of the sections were in Banaskanta (Palanpur) district of Gujarat.

The sections of Gandhidham Area were: Gandhidham (GIM)-Palanpur (PNU) (Meter Gauge (MG)), Gandhidham (GIMB)–Samakhiyali (SIOB) (Broad Gauge (BG)), GIMB–New Bhuj (NBVJ) (BG), Adipur (AI)-Mundra Port (MDPT) (BG), and Gandhidham (GIM/GIMB)-Kandla Port (KDLP) (MG/BG) section.

About 18 BG mail/express and passenger trains, and 10 MG passenger trains besides one Rail Bus originated from Gandhidham Area. About 18 to 20 goods trains ran per day in the Gandhidham Area.

The train operations were controlled by one control office situated at Gandhidham. The control office had interchange points at Samakhiyali on the BG section with the Ahmedabad control office, at Bhildi on the MG section with Jodhpur Division of North Western Railway, and at Palanpur with Ajmer Division of WR.

The sectionwise details of Gandhidham Area are:

Section	Station Class			Length (km)
	“B”	“D”	SPL	
GIM-PNU (MG)*	20	15	0	300.81
GIMB-SIOB (BG)	7	0	0	52.98
GIMB-NBVJ	4	4	0	58.25
NBVJ–NLY	0	9	1	101.40
KDLP-GIMB	2	0	0	11.60
AI-MDPT	3 [#]	0	0	56.24

* GIM-PNU Section, excluding PNU

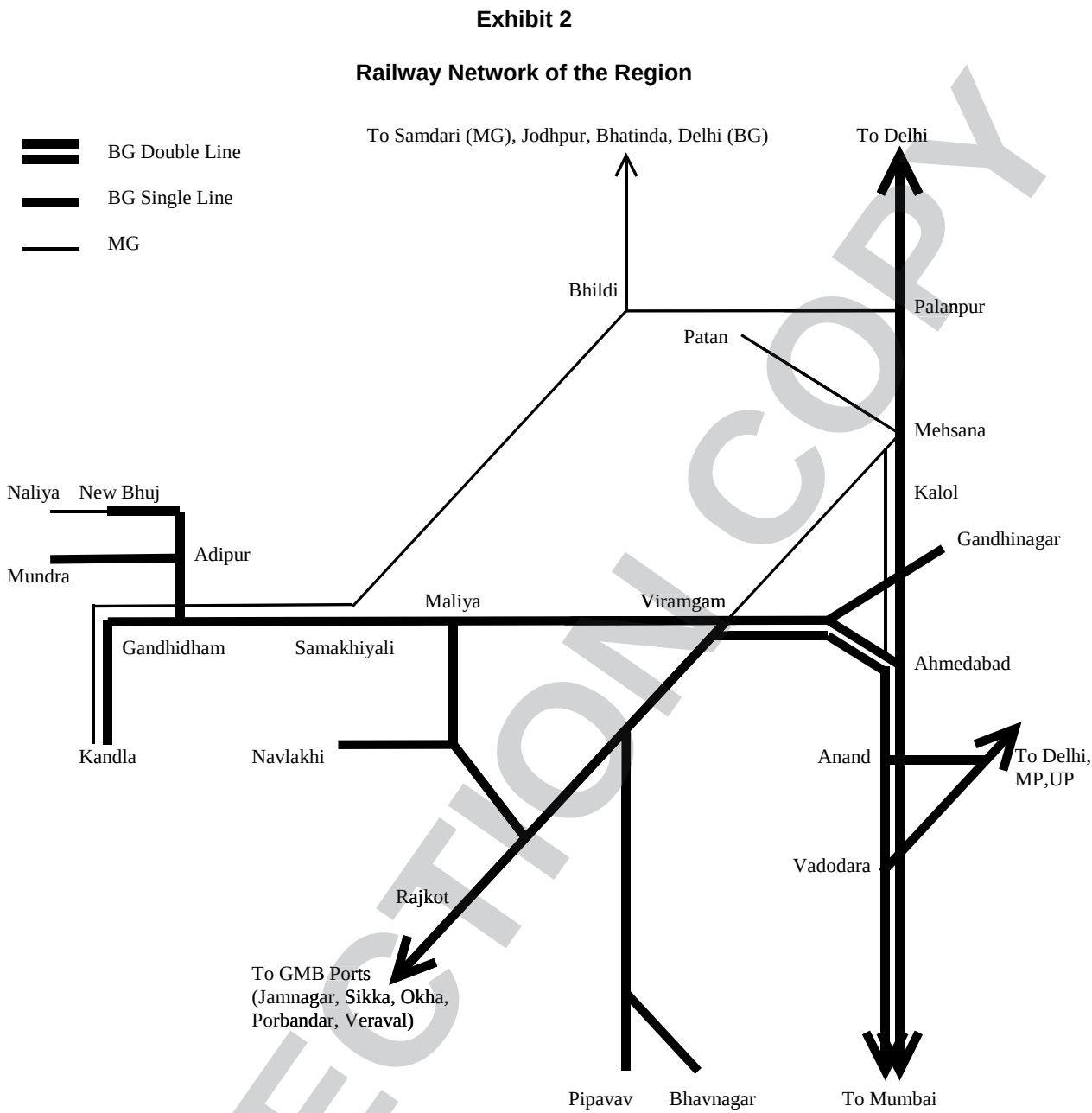
Excluding Bhadreswar “B” class Station of AI-MDPT section (station not opened)

Some Milestones in the History of Gandhidham Area

- The importance of the entire area received tremendous impetus after partition in 1947 when Karachi port went to Pakistan. Because of its geographical contiguity and the possibility of developing Kandla port to serve as a hinterland of Rajasthan, Punjab, Haryana, and Delhi and Western UP, in September 1948, the Railway Board was asked to provide MG and BG connections to Kandla.
- The construction of the MG line was started in January 1950 and opened to traffic on 2nd October 1952 by the then President of India, Dr Rajendra Prasad. The BG link from Viramgam to Kandla was established in 1969.

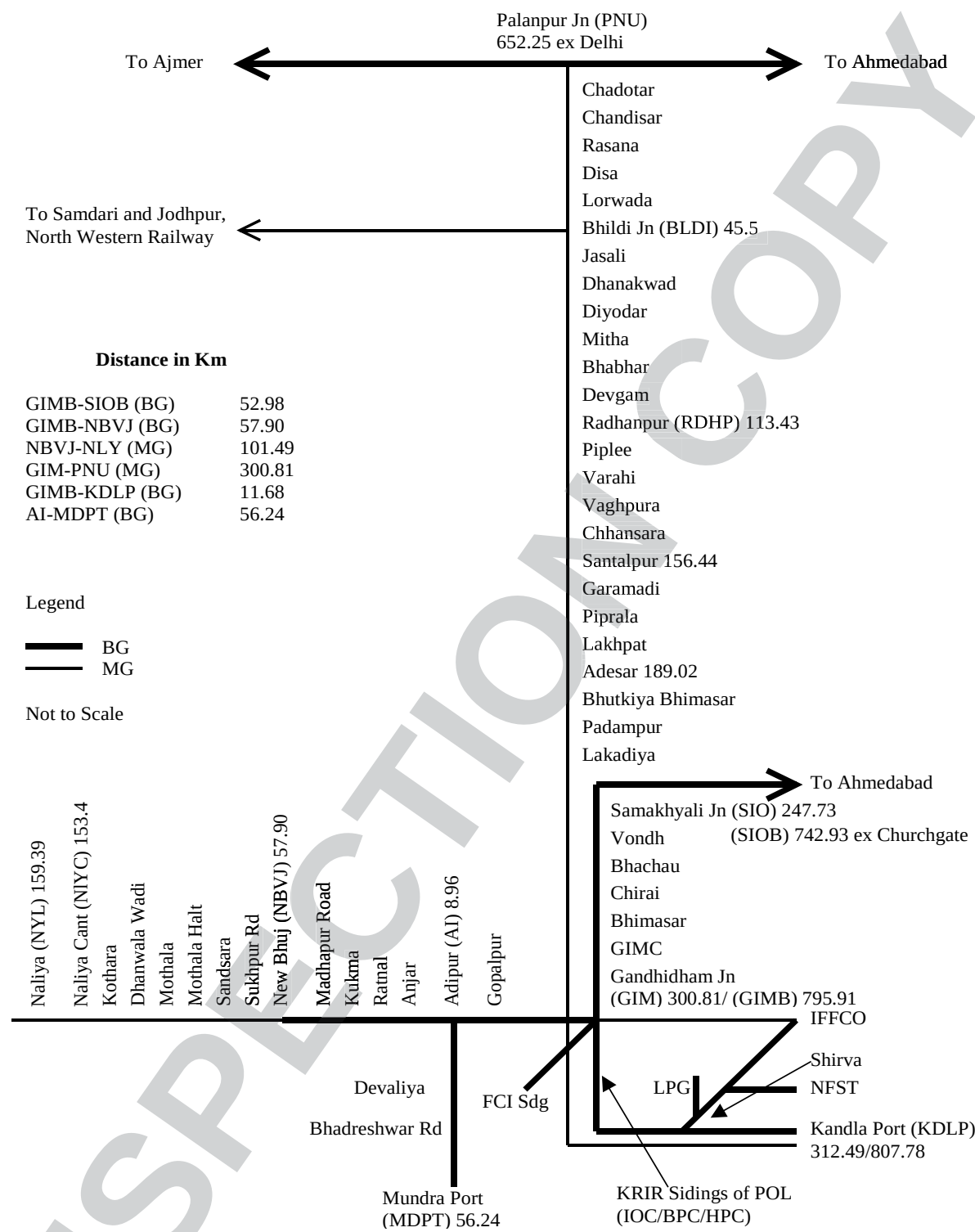
- The earlier Kutch State Railway's Narrow Gauge (NG) line from Bhuj to Adipur was converted into MG and linked to Gandhidham in 1954.
- On 08.08.1990, New Bhuj–Naliya MG section was opened due to its strategic importance.
- On 08.11.2000, Gandhidham–New Bhuj Section was blocked for MG to BG gauge conversion. On 15.05.2001, it was opened for traffic. On 03.06.2001, Kutch Express to Mumbai was flagged off by Hon'ble Prime Minister Shri Atal Bihari Vajpayee from New Bhuj.
- From 15.11.2001, Adipur-Mundra port private BG rail link was opened for goods traffic only.
- Previously, the Gandhidham Area was part of Ajmer Division of WR. From 01.04.2003, it was merged into the newly opened Ahmedabad Division of WR.

Source: Area Railway Manager, Gandhidham, Western Railway, 2004



Source: Map of the Railway Time Table, Western Railway, 2003

Exhibit 3: Layout of Gandhidham Area



Source: Area Railway Manager, Gandhidham, Western Railway, 2004

Exhibit 4

Stationwise Goods Handling in Gandhidham Area

Handling Station /Siding	Code		Commodities
Gandhidham Goods Shed	MG	GIM	Salt, imported fertilizer, container, RMC, edible oil, food grains, military store, sugar etc. Except POL
	BG	GIMB	
Kandla Port	BG	KDLP	Imported fertilizer, food grains, coal, container, crude edible oil, HR coils, scrap iron etc. Except POL and crane consignment
Chirai (outward only)	BG	CHIB	Salt
	MG	CHII	
Bhimasar	BG	BMSB	Salt and coal. Except POL and crane consignment
	MG	BMSR	
Samakhiyali	BG	SIOB	Salt
	MG	SIO	
Santalpur	MG	SNLR	Salt
Adesar	MG	AAR	Salt
NFST Siding (IOC)	BG	NFST	POL/Naphtha/Lubricant Only
FCI Siding	MG	FIG	Food grains
	BG		

Source: Area Railway Manager, Gandhidham, Western Railway, 2004

Exhibit 5
Major Ports of India



Source: www.mapsofindia.com, accessed on August 11, 2004 and Authors' Modifications, 2004

Exhibit 6

Exports and Rail Share at Kandla Port

	2003-04		2002-03		2001-02		2000-01		1999-00	
Commodity	Total Qty tons	Rail %	Total Qty tons	Rail %	Total Qty tons	Rail %	Total Qty tons	Rail %	Total Qty Tons	Rail %
Wheat	2715075	96.27	2049603	83.16	1027539	77.20	495881	27.26		0.00
Rice	1960772	8.14	3384022	67.39	1531364	57.34	679796	18.29	843421	0.28
Container	1604615	1.13	1542314	0.17	1163799	0.16	805459	0.28	703066	0.14
DOC	1592434	39.43	986780	26.68	1585999	55.34	1139673	62.50	1048551	56.73
Salt	862856	0.00	927951	0.00	1298594	0.00	834859	0.00	403494	0.00
POL	340000	101.01	415808	102.59	210508	95.10	37578	76.51	0	0.00
Zinc Concentrate	255123	0.00	155744	0.00	53710	0.00	0	0.00	0	0.00
Edible/ Castor Oil	239950	0.00	159698	0.00	178885	0.00	178778	0.00	190885	0.00
Ores including Bentonite	145415	0.00	156382	0.00	164874	0.00	237309	0.00	255130	0.00
Maize/ Bajra/ Jowar	126886	18.11	8074	0.00	17542	13.35	11001	0.00	0	0.00
Iron and Steel/Pipe	28931	0.00	56350	0.00	30059	0.00	9585	0.00	53145	0.00
Sesame Seed	12031	0.00	13717	0.00	27405	0.00	51885	0.00	16880	0.00
Others	8906	0.00	21074	0.00	13184	0.00	41855	14.56	4292	0.00
Total	10308182	36.81	10378897	45.27	7726921	35.81	4777042	21.12	4027892	14.85

- Total quantity is the quantity handled at the port, based on all hinterland modes.
- The quantity excludes transshipment traffic.

Source: Area Railway Manager, Gandhidham, Western Railway, 2004

Exhibit 7

Imports and Rail Share at Kandla Port

	2003-04		2002-03		2001-02		2000-01		1999-00	
Commodity	Total Qty Tons	Rail %	Total Qty tons	Rail %	Total Qty tons	Rail %	Total Qty tons	Rail %	Total Qty tons	Rail %
POL	1582236	6.28	1111927	6.59	892126	19.26	2251321	16.05	9993787	15.86
Timber Logs	1427780	0.00	1398588	0.00	1206450	0.00	1298375	0.00	1155143	0.00
Container	801291	2.44	689191	0.88	586226	1.40	491426	0.12	431164	0.16
Iron Scrap	416759	0.00	385167	0.00	727991	0.00	536555	0.00	526161	0.00
MOP	396165	22.5 7	357697	11.69	269700	11.07	252797	11.70	370438	17.41
Edible Oil	338524	16.6 1	307085	17.05	354298	9.00	389442	0.32	812632	0.00
Sulphur	268273	0.00	292619	0.00	237332	0.00	229184	0.00	370506	0.00
Coal	243881	7.13	254475	27.16	335009	0.00	624356	4.39	828998	27.40
HR Coil	194114	0.00	161673	35.24	0	0.00	0	0.00	0	0.00
DAP/Urea	105909	75.0 3	157962	73.47	341042	78.48	219360	89.38	649713	96.22
Ores including Zinc Concentrate	28420	0.00	15000	0.00	36382	0.00	47814	0.00	58580	0.00
Food Grains	669	0.00	638	0.00	39	0.00	85939	5.31	205492	0.00
Rock Phosphate	0	0.00	0	0.00	27500	19.18	0	0.00	125042	58.22
Sugar	0	0.00	0	0.00	0	0.00	0	0.00	97066	13.86
Others including IFFCO Liquid Fertilizer Raw Material	3708887		3622928		3455359		2720998		2392431	
Total (Kandla)	9512908	3.80	8754950	4.13	8469454	6.08	9147567	6.79	18017153	14.37
IFFCO Output Traffic Handled by Rail	2005773		2257536		1938708		1633231		1816055	
Rail % including IFFCO		24.8 8		29.92		28.97		24.64		24.45
Import at Vadinar	21566909		20929374		19947047		18830680		18078126	
Total (Kandla + Vadinar)	31079817		29684324		28416501		27978247		36095279	

- Kandla port has two locations handling imports: Kandla and Vadinar. Vadinar handles only crude oil imports, which are evacuated by pipeline.
- Raw material (liquid) for IFFCO plant, located adjacent to the port, is handled by pipeline to the plant. This is part of the total traffic handled at Kandla.
- “Total Qty” is the quantity handled by the port, based on all hinterland modes. The quantity excludes transshipment traffic.

Source: Area Railway Manager, Gandhidham, Western Railway, 2004

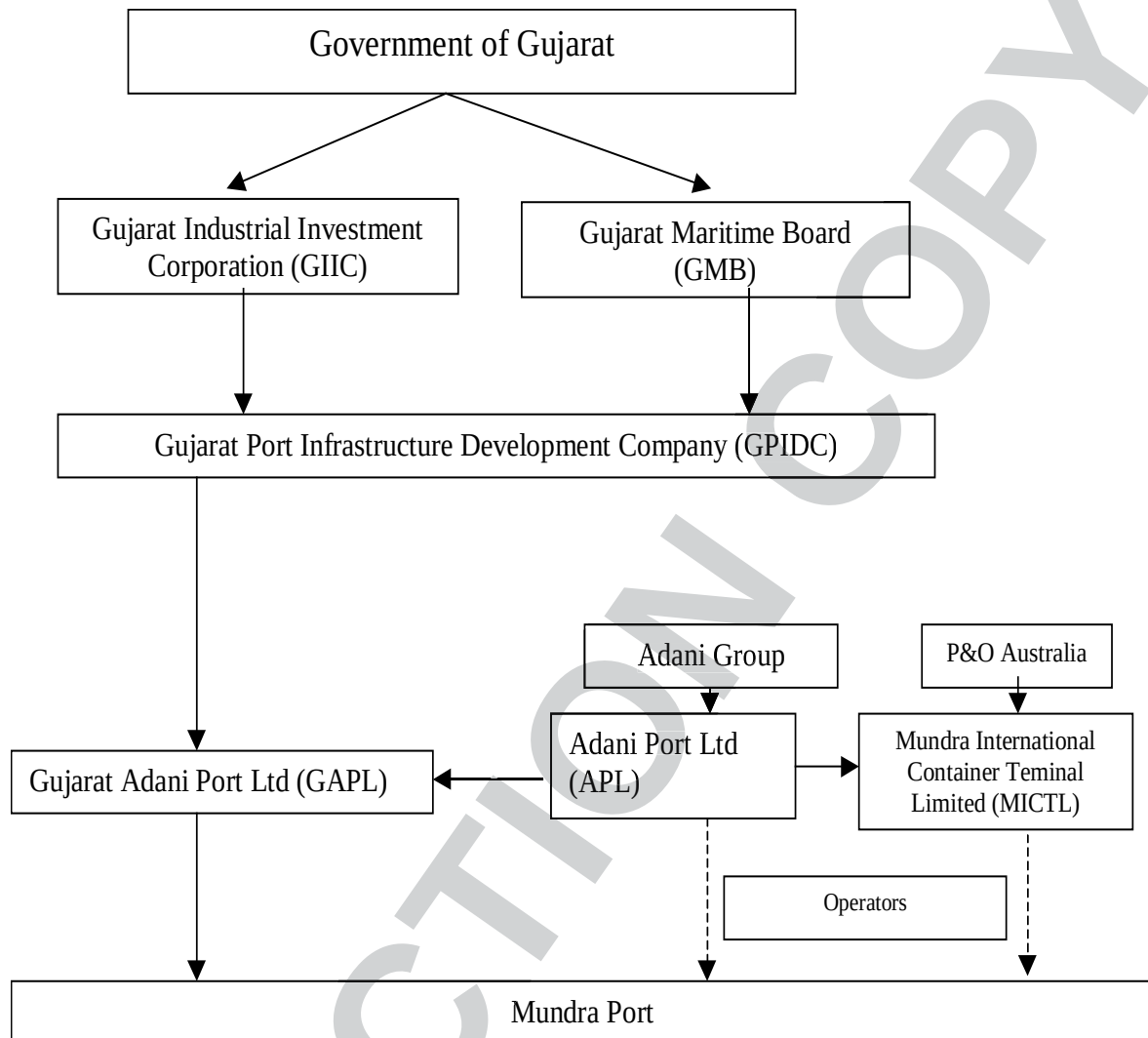
Exhibit 8

Hinterland Distance Profile at Kandla Port

Rank	Commodity		Quantity tons	Share of Total %	Cumulative Share %	ton km	Average Hinterland Distance km
Top 10 Liquid Commodities for November, 2003							
1	Phos Acid	Import	152939	38.70	38.70	0	0.00
2	CPO-E/Grade	Import	31927	8.08	46.78	40072937	1255.14
3	Furnace Oil	Import	23922	6.05	52.83	6297000	263.23
4	Ammonia	Import	19531	4.94	57.77	0	0.00
5	SKO	Import	19412	4.91	62.68	18172302	936.14
6	Naptha	Export	19200	4.86	67.54	0	0.00
7	Butane	Import	17463	4.42	71.96	0	0.00
8	CPO-I/Grade	Import	14494	3.67	75.63	15141398	1044.67
9	Methanol	Import	14018	3.55	79.18	15392570	1098.06
10	Castor Oil	Export	10242	2.59	81.77	7813470	762.89
	Total		323148	81.77			
	Others		72057	18.23			
	Grand Total		395205	100.00			
Top 10 Liquid Commodities for February, 2004							
1	Phos Acid	Import	107775	25.72	25.72	0	0.00
2	FO	Import	45085	10.76	36.49	252000	5.59
3	CPO	Import	40135	9.58	46.07	43584648	1085.95
4	Ammonia	Import	38875	9.28	55.34	0	0.00
5	Butane	Import	36644	8.75	64.09	0	0.00
6	Naphtha	Both	29516	7.05	71.14	2238384	75.84
7	HSD	Import	10751	2.57	73.70	5508160	512.34
8	Castor Oil	Export	8680	2.07	75.77	6349000	731.45
9	Crude Palm Oil	Import	7812	1.86	77.64	8836717	1131.17
10	Benzene	Export	7635	1.82	79.46	6947850	910.00
	Total		332908	79.46			
	Others		86049	20.54			
	Grand Total		418957	100.00			

Source: Kandla Port and Authors' Analysis, 2004

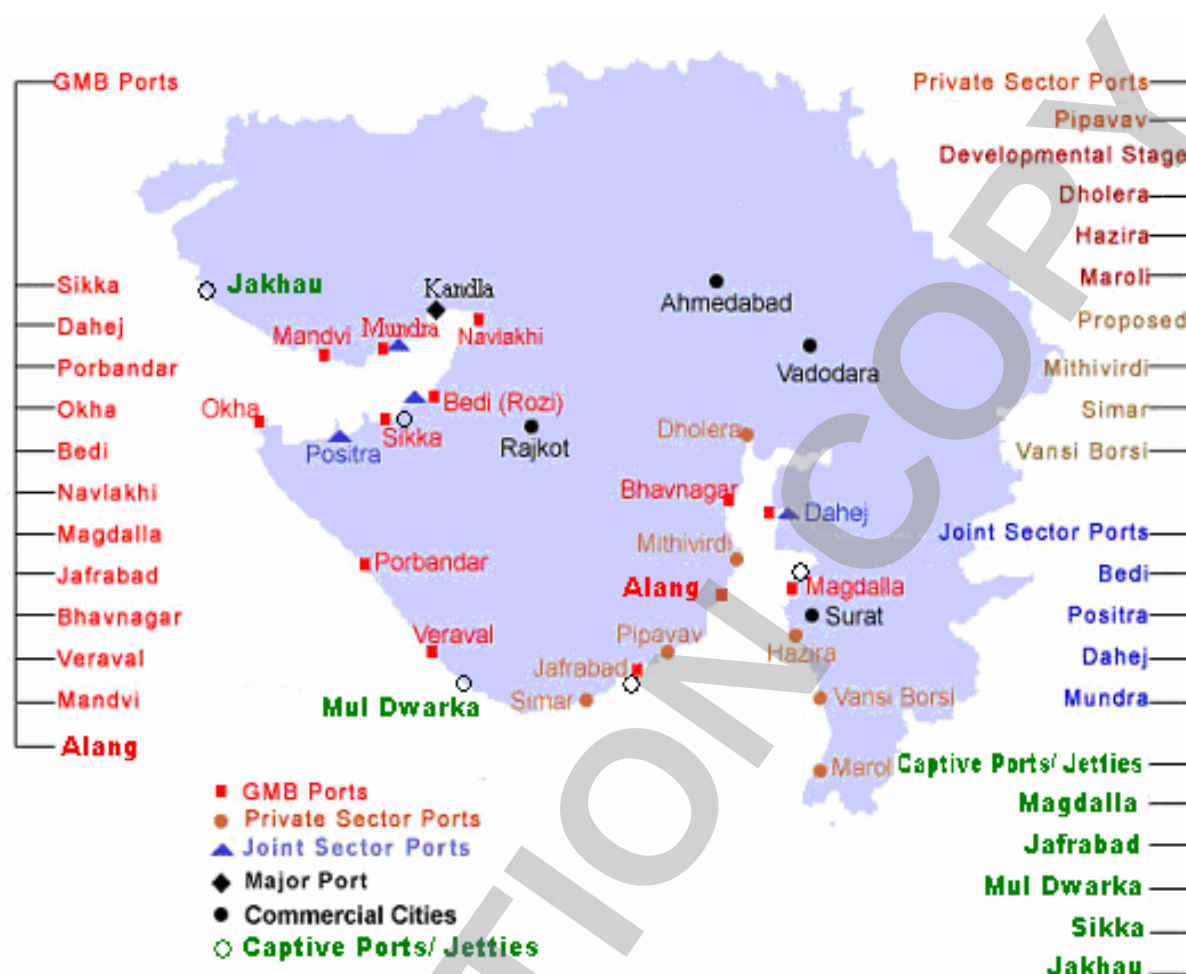
Exhibit 9
Ownership Structure of Mundra Port



- Railway Siding from Adipur to Mundra port was developed and operated by a subsidiary called Gujarat Adani Infrastructure Ltd.
- The group is partnering an SPV, along with Kandla Port and Indian Railways, called Kutchch Railway Corporation Ltd, for gauge conversion projects.

Source: www.mundraport.com , accessed on August 11, 2004 and Authors' Analysis, 2004

Exhibit 10: Ports in Gujarat



Source: www.gmbports.org, accessed on August 14, 2004 and Authors' Modifications, 2004

Exhibit 11
Mundra Port Cargo Since Inception

(tons)

	1998-99	1999-00	2000-01	2001-02	2002-03	2003-04 (Until Feb)	Total
Dry Cargo							
Fertilizer		116569	128600	282168	131728	313639	972704
Coal		204907	781403	893408	1045923	521215	3446856
Steel		3296	60543	334264	493557	548377	1440037
Wheat		13000	81000	535330	662766	572888	1864984
Rice	13500				310447	54399	378346
DOC	99646	149636	152917	133270	60182	116588	712239
Clinkers					168899	352103	521002
Minerals	9238	209585	389177	357600	413184	604690	1983474
Others	28066	143267	154751	48473	27839	47090	449486
Project Cargo			12221		4101	3344	19666
Total Dry	150450	840260	1760612	2584513	3318626	3134333	11788794
Liquid Cargo							
Vegetable Oil	54198	226032	459724	488526	574805	566612	2369897
Chemical		79569	136633	256621	216260	216451	905534
POL	39445	50618	131976	138544	94231	216737	671551
Total Liquid	93643	356219	728333	883691	885296	999800	3946982
Grand Total (Dry + Liquid)	244093	1196479	2488945	3468204	4203922	4134133	15735776

Source: Mundra Port, 2004

Exhibit 12

Inward Rail Cargo at Mundra (Export)

Product	Nov 01-Mar 02			Apr 02-Mar 03			Apr 03-Feb 04			Nov 01-Feb 04		
	Rakes	Wt tons	Freight Rs m	Rakes	Wt tons	Freight Rs m	Rakes	Wt tons	Freight Rs m	Rakes	Wt tons	Freight Rs m
Wheat	96	222414	110	291	678746	420	239	557050	420	626	1458210	950
DOC	18	34390	20	2	3413	0	32	60598	40	52	98401	60
Rice	1	2342	0	100	228683	180	8	18745	10	109	249770	192
Container	0	0	0	1	0	0	83	0	0	84	0	0
Total	115	259146	130	394	910842	600	362	636393	470	871	1806381	1200

Source: Mundra Port, 2004

Exhibit 13

Outward Rail Cargo at Mundra (Import)

Product	Nov 01-Mar 02			Apr 02-Mar 03			Apr 03-Feb 04			Nov 01-Feb 04		
	Rake S	Wt Tons	Freight Rs m	Rake S	Wt tons	Freight Rs m	Rake S	Wt tons	Freight Rs m	Rake S	Wt Tons	Freight Rs m
Coal	87	298898	100	287	975783	320	121	418951	130	495	1693632	551
Fertiliser	29	66962	20	49	114031	80	112	261174	190	190	442167	301
Chick Peas	1	2342	0	0	0	0	1	2345	0	2	4687	4
Vegetable Oil	3	1562	0	26	14900	20	33	25688	30	62	42150	55
Wheat	0	0	0	2	3978	0	2	4620	0	4	8598	1
Salt/M-Coke	0	0	0	0	0	0	2	4342	10	2	4342	7
Container	0	0	0	1	0	0	58	0	0	59	0	0
Total	120	369764	130	365	1108692	420	329	717120	360	814	2195576	920

Source: Mundra Port, 2004

Exhibit 14

Exports and Rail Share at Mundra Port

Commodity	2003-04		2002-03		2001-02	
	Qty tons	Rail %	Qty tons	Rail %	Qty tons	Rail %
Wheat	694311	85.23	1020406*	89.59	540414*	62.10
Bauxite	485971	0	144141	0	194149	0
Clinker	352102	0	179074	0		
Bentonite	181201	0	157137	0	144469	0
Pipes	134290	0	201579	0	127903	0
SBM(DOC)	114553	63.83	5880	58.04	154406	35.74
Bentonite (in bags)	97504	0	45982	0	26967	0
Rice	60853	30.80				
Rape Seed	31841	0	20264	0		
Salt	25748	0	22214	0	16240	0
Castor Seed	23989	0	18603.5	0		
Maize	10055	0				
Clay Granules					1665	0
Total	2212418	30.90	1776413	51.66	1206213	32.40
* Foodgrains including wheat, rice, etc						

Source: Area Railway Manager, Gandhidham, Western Railway, 2004.

Exhibit 15

Imports and Rail Share at Mundra Port

Commodity	2003-04		2002-03		2001-02	
	Qty tons	Rail %	Qty tons	Rail %	Qty tons	Rail %
Coal	693691	70.80	1048522	89.97	1065118	83.25
Edible Oil	523969	4.56	566463	3.04	466641	2.71
Plates/HR Coil	417667	0	250935	0	91118	0
DAP/Urea	201069	64.22	6020	0	152774	84.89
MOP	176394	44.59	125707	68.39	95311	31.70
Flourspar and Ammonium Nitrate	14289	0			2608	0
Peas/Chickpeas	9336	25.12			19652	11.87
Timber Logs	8772	0				
Bitumen Drums	550	0			6488	0
Scrap	180	0	16390	0		
Iron Ore			5725	0		
Sulphur					18374	0
Total	2045917	35.44	2014037	51.96	1918084	55.35

Source: Area Railway Manager, Gandhidham, Western Railway, 2004.

Exhibit 16

Rail Tariff for Major Commodities from Gandhidham

(Rs per quintal)

Destination	Distance km	Load	Wheat	Rice	Power House Coal	Timber Sawn	Shredded Steel/ Scrap	Ground Nut Oil in Tins
Kankaria	302	Train	17.9	17.9	25.8	25.8	31.8	17.9
		Wagon	18.9	18.9	26.8	27.8	35.8	18.9
Surat	529	Train	28.7	28.7	41.5	41.5	51.1	28.7
		Wagon	30.3	30.3	43.1	44.7	57.5	30.3
Alwar	1064	Train	55.7	55.7	80.5	80.5	99.1	55.7
		Wagon	58.8	58.8	83.6	86.7	111.5	58.8
Patiala	1532	Train	79.0	79.0	114.1	114.1	140.5	79.0
		Wagon	83.4	83.4	118.5	122.9	158.1	83.4

Source: FOIS, Indian Railways, 2004.

Exhibit 17

Traffic Projection of Gandhidham Area

Port	2003-04			2006-07			2011-12		
	Total Traffic	By Rail		Total Traffic	By Rail		Total Traffic	By Rail	
	mt/no	mt/no	Rakes	mt/no	mt/no	Rakes	mt/no	mt/no	Rakes
Export									
Kandla	5.49	2.47	1051	7.44	3.35	1425	11.89	5.35	2277
Mundra Adani	2.00	1.20	510	3.60	2.16	919	6.20	3.72	1583
Mundra Container (TEUs)	20000	12000	133	80000	48000	533	220000	132000	1467
Total Export Rakes			1694			2877			5327
Average/Day			4.64			7.88			14.59
Import									
Kandla	11.25	3.37	1434	13.89	4.17	1774	19.47	5.84	2485
Mundra Adani	3.01	1.81	770	5.50	3.30	1404	9.10	5.46	2323
Mundra Container (TEUs)	20000	12000	133	80000	48000	533	220000	132000	1467
Salt		01.0	425		1.5	638		2.0	851
Total Import Rakes			2762			4349			7126
Average/Day			7.57			11.92			19.52
Total Rakes			4456			7226			12453
Average/Day			14.21			19.80			34.11
Empty Rakes 30%			2.85			4.0			6.80
Overall Average/Day			17.00			24.00			41.00
Average Trains in Each Direction			8.5			12.00			20.50

- Kandla Port traffic handled by rail includes IFFCO, GIMB, NFST, KRIR, LPGS and KDLP
- Total production of salt included in export by both ports, rest handled by rail/road
- For Kandla port projections, 30% of import and 45% of export traffic handled by rail.
- For Mundra Port projections, 60% of total traffic handled by rail as per projections given by Mundra Adani Port.

Source: AF Ferguson Report for Mundra Port and internal traffic projection for Kandla Port

Exhibit 18

Existing Rake Handling Capacity

S. No.	Station	Rake Handling Capacity
1	Gandhidham	4.5 + 1 in new siding
2	Kandla Port	3.5 outside + 3 inside + 1 coal/container
3	Mundra Port	7 (3 cargo + 1 container + 2 coal) + 1 outside
4	Chirai	1 salt
5	Bhimasar	1 salt
6	Samakhiyali	1 salt
7	Bhuj	1
8	IFFCO Siding	4
9	NFST	2
10	LPG Siding (IOC)	2
11	POL Sidings – GIM	6

Source: Area Railway Manager, Gandhidham, Western Railway, 2004.

Exhibit 19

Present Line Capacity of Gandhidham-Maliya Section

Track capacity utilized by passenger trains and margin left over for goods trains						
Year	Capacity	Trains Run			Utilization	Margin
		Passenger	Goods	Total	Passenger Share %	Left Over for Goods Trains
2001-02	15.0	3.6	7.4	11.0	24.0	11.4
2002-03	15.5	5.7	9.1	14.8	38.0	9.8

Source: Area Railway Manager, Gandhidham, Western Railway, 2004.

Exhibit 20

Gujarat Maritime Board - Vision

- Each port of GMB to be established as 'THE PORT OF CHOICE'.
- 100 per cent automation of cargo- handling process through mechanization and/or instrumentation and control technology.
- Fully computerized systems
- Adequate berthing and storage facilities at ports
- Pre-berthing delay non-existent and average turnaround time nearing international standards for direct berthing ports.
- Each port acting as an independent profit centre.
- Commodity group-specific port terminals with optimal synergy and focus.
- Extensive linkages of ports with hinterlands along with sufficient infrastructure backup.

Source: www.gmbports.org, accessed on August 17, 2004

Exhibit 21

Gujarat Maritime Board - Major Growth Commodities
mt

Commodity	2002-03 (Actuals)	2007-08	2012-13	2017-18	Basis
Crude Oil	27.74	76.00	112.00	142.00	Growth in crude oil traffic by Reliance Refinery. Future developments at Mundra, Dahej and Hazira.
LNG	0.00	12.00	21.00	29.00	LNG terminals at Dahej and Hazira reaching their peak capacities
Coal	7.69	13.00	17.75	22.00	Consumption growth from power plants, cement sector and steel sector
Iron Ore/Enriched Iron Ore	3.42	8.40	12.40	16.40	As per the growth plans of Essar Steel
Cement/Clinker	9.73	13.50	17.80	22.00	Average growth of 8% - execution of road infrastructure projects
Containers	0.01	13.20	35.99	63.90	Container terminals at Mundra and Pipavav in operation. Container terminal at Hazira expected to be operational before 2008
Total	48.59	136.10	216.94	295.30	

Source: Gujarat Maritime Board, Annual Report 2004

Exhibit 22

Status of Works in Gandhidham and Kandla Goods Shed

S. No.	Particulars of Work	Cost (Rs m)	Status of work		Whether Contract awarded	Target
			Estimate	Plan		
1	Gandhidham Goods Shed					
(A)	Sanctioned work					
1	Improvement to goods shed (ADI (U/02-03))	2.97	Sanctioned	Approved	Yes	30-11-2003
2	Provision of additional line at goods shed	2.99	Sanctioned	Approved	Yes	30-11-2003
3	Improvement to goods shed area (2003-04)	5.90	Under Process	Under Process	-	(I) NIT being issued pending sanction of detailed estimated. (II) Target Jun-04
(B)	Q. Additional work proposed					
4	Providing WBM and asphalting in goods line no 1 area to facilitate truck movement	2.95	-	-	-	Work being proposed on an out of turn basis.
5	Providing WBM adjoining line no 4 to facilitate truck movement.	2.90	-	-	-	Work being proposed on an out of turn basis.
2	Kandla Goods Shed					
(A)	R. Sanctioned work					
1	Kandla port: Providing rail level loading platform for goods terminal at line no 6 (ADI/U/02-03)	2.96	Sanctioned	Approved	Yes	To be completed by 15-10-2003
2	Kandla port: Providing full rake track siding at goods terminal	2.98	Sanctioned	Approved	Yes	Completed
3	Kandla Port BG full rake handling facility	16.56	-	-	-	Work being done by construction department
(B)	S. Additional work proposed					
4	Improvement to Kandla goods area by providing WBM and asphalting to facilitate truck movement	2.95	-	-	-	Work being processed on out of turn basis

Source: Area Railway Manager, Gandhidham, Western Railway, 2004

Exhibit 23

Future Projects in the Gandhidham Area

S. No.	Project	Sanctioned	Proposed	Not Yet Proposed
1	Gauge conversion of SIOB-PNU section (being undertaken by an SPV called Kutchch Railway Corporation Ltd)	Y		
2	Double line between SIOB-GIM section	Y		
3	Double line between GIM BG Yard-KDLP section		Y	
4	Bypass chord to MDPT at GIMB	Y		
5	Conversion of Gandhidham block station into full fledged block station with two loops. (One loop as goods shed)		Y	
6	Separate salt loading siding at CHIB		Y	
7	Bypass chord to SIOB-LKZ during conversion of SIOB-PNU section		Y	
8	One additional loop line at VONB to work as Goods shed		Y	
9	One additional loop line at AI		Y	
10	Additional Block station between CHIB-BCOB			Y
11	Expansion of GIM yard. Conversion of existing MG yard into BG			Y
12	C&W examination facility at Mundra Port			Y
13	Increase in C&W facility at GIM			Y
14	Expansion of KDLP yard with 6 rakes handling inside the port and six rakes handling outside the port		Y	
15	Expansion of GIMB good shed to nine rakes with six rakes at good shed, two rakes at Khari siding and one at GIM bypass line			Y
16	Covered good shed at GIM and KDLP			Y
17	Two good shed lines at BMSR			Y
18	Two good shed lines at CHIB		Y	
19	Separate salt loading siding at VONB		Y	
20	Separate salt loading siding at SNLR for one rake and AAR for one rake, during gauge conversion	Y		
21	Expansion in existing rake handling facility at MDPT by three rakes (two for cargo, one for container)			Y

Source: Area Railway Manager, Gandhidham, Western Railway, 2004